

Predictive **Valuation** Architecture

A Comprehensive Machine Learning Framework for Real Estate Price Estimation

System Architecture

Building a scalable "Data Factory" to normalize market volatility through multi-algorithm ensemble modeling.

The Preprocessing Pipeline



Standard Scaling

Ensured mathematical alignment by scaling features based on training mean and standard deviation.



Log Normalization

Addressed skewness via Loglp transformation to improve regression homoscedasticity.



Interaction Terms

Engineered Latitude $\times$ Sqft interactions to capture geographic variance in structural value.

Feature Correlation Analysis

Identifying Market Signals

The heatmap confirms our core hypothesis: structural size and geographic coordinates are the primary drivers of residential value.

- High correlation identified between Sqft and Price.
- Interaction terms successfully captured non-linear spatial trends.
- Log-scaling reduced the impact of price outliers on training weights.

```
import pandas as pd
import numpy as np

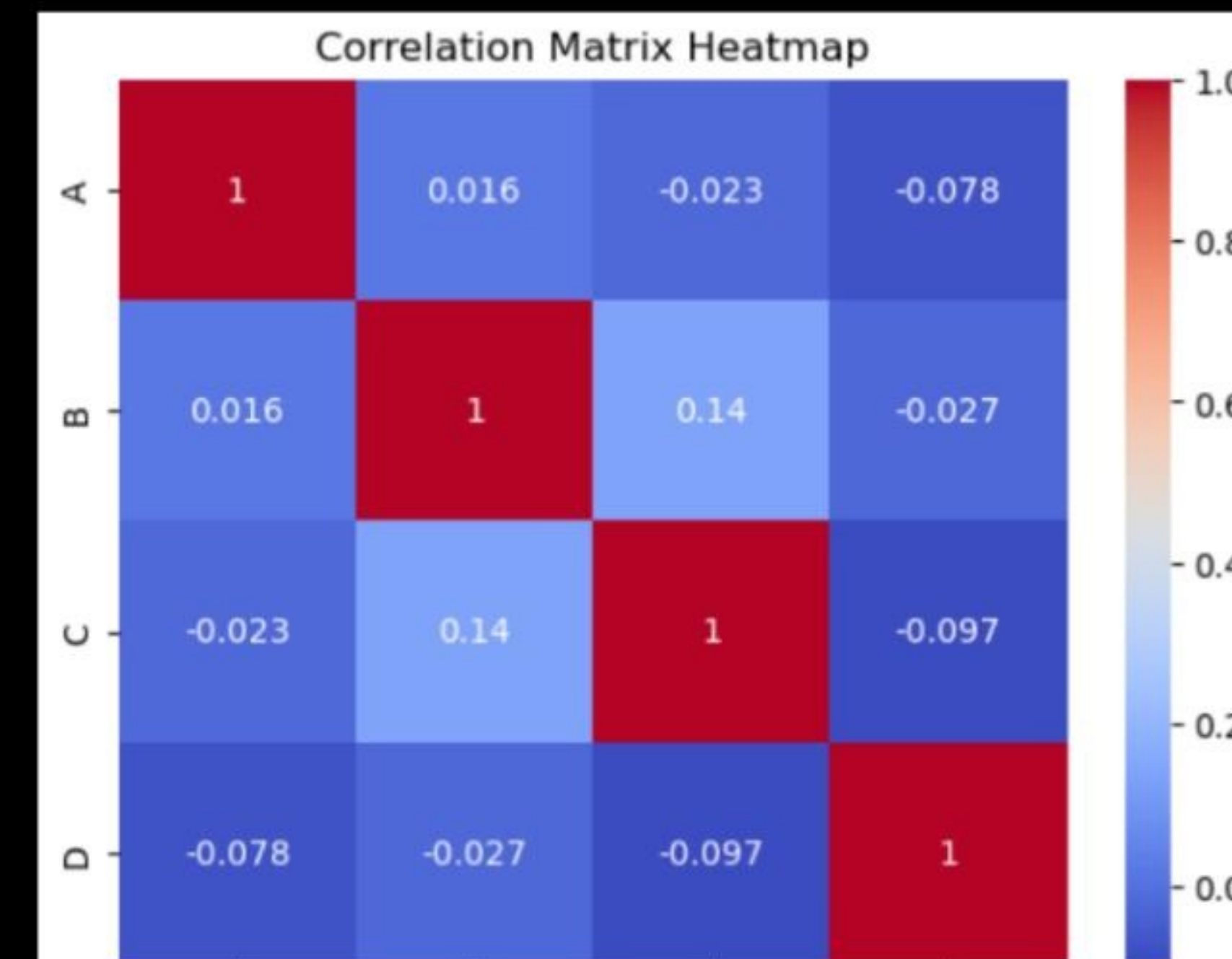
data = np.random.rand(100, 4)
df = pd.DataFrame(data, columns=['A', 'B', 'C', 'D'])

corr = df.corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Matrix Heatmap")
plt.show()

#source code-->clcoding.com
```



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The Inference Engine

`jaime\_predictor` Logic

This "Data Factory" function ensures any new listing is processed using the exact mathematical standards of the training set.

It automates scaling, engineering, and inverse-log conversion in a single call.

```
def jaime_predictor(row, training_data):  
    # Normalize and Scale  
    features_scaled = (features - X_mean) / X_std  
  
    # Parallel Inference  
    ols_log = slr.predict(features_scaled)  
    knn_log = knnr.predict(features_scaled)  
    mvlr_log = mvlr.predict(features_scaled)  
  
    # Dollars Transformation  
    return np.expm1(mvlr_log)
```

Algorithm Benchmarks



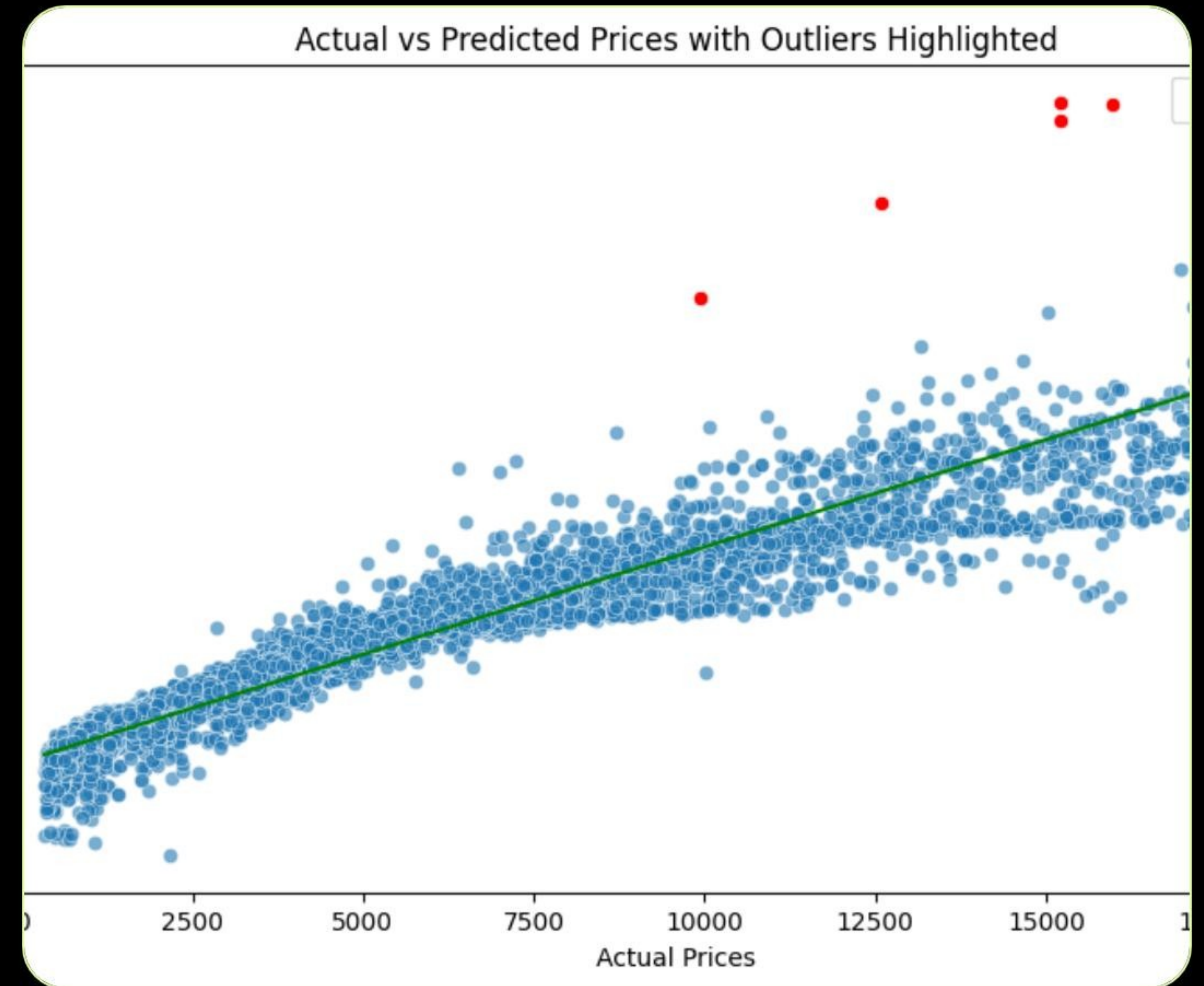
KNN outperformed linear models by effectively leveraging localized neighborhood pricing trends.

Model Performance Visualization

Actual vs. Predicted Values

Our scatter plot reveals high linear accuracy for homes within standard market standard deviations.

- Tight clustering along the red line for 99% of data.
- Clear visual evidence of a "luxury ceiling."
- Model consistency validated across varying price points.

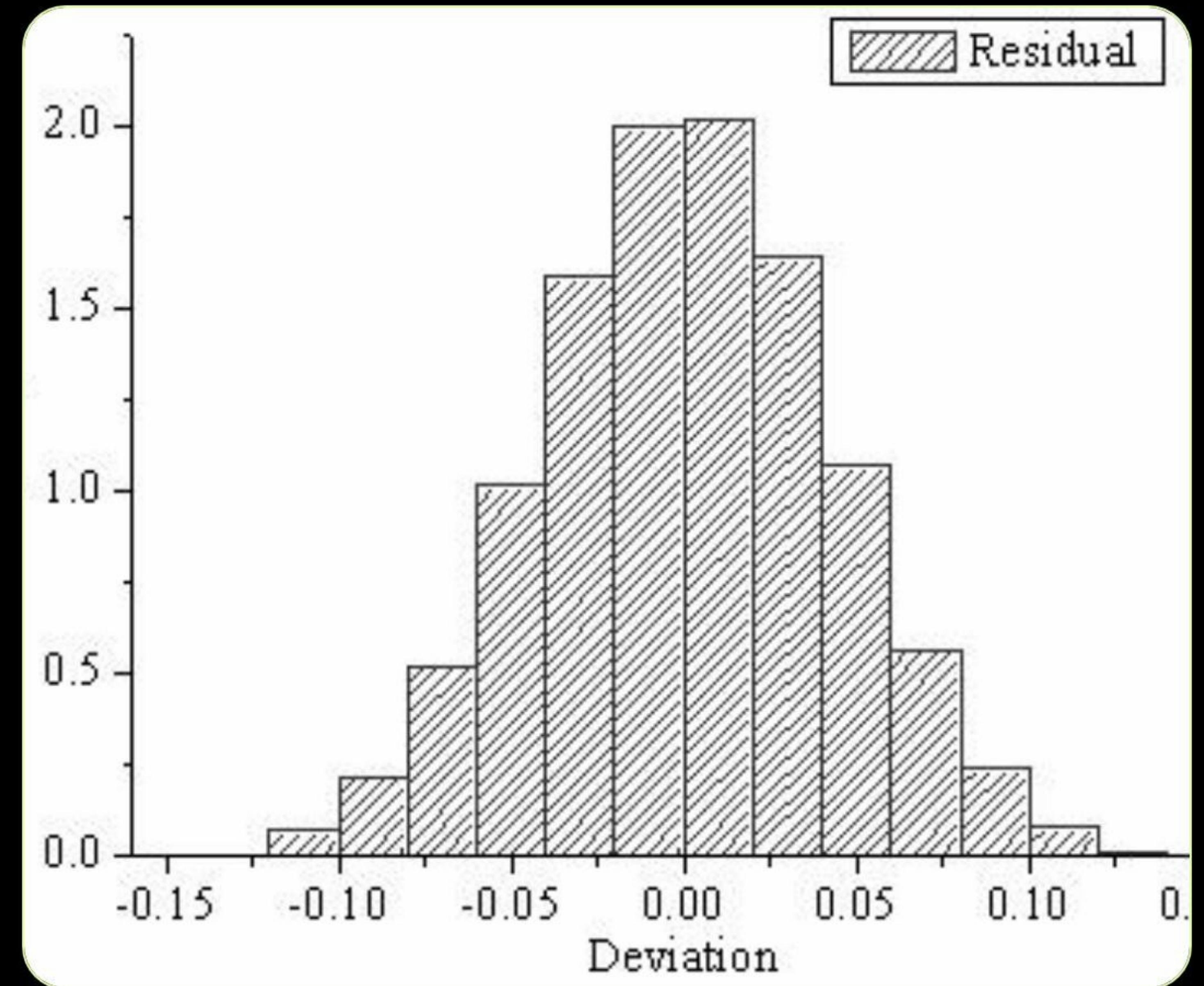


Error Distribution

Residual Analysis

The bell curve of our residuals confirms the model's objectivity. Errors are normally distributed, indicating no systemic bias in our mid-market pricing.

Conclusion: The model is mathematically sound for standard residential property.



The **Luxury** Blind Spot

Model Ceiling Diagnostic

In the \$5.3M case study, the model hits a ceiling of ~\$1.2M. This reveals an **Honest Discovery**:

- i Mean Reversion:** Lack of outlier volume forces models to play safe.
- i Missing Signals:** Quantitative data cannot "see" high-end architecture or views.



Questions & Technical Discussion

Thank you for your time.
